MAPÚA MALAYAN DIGITAL COLLEGE
of Mapúa Malayan Colleges Laguna
In collaboration with
Arizona State University®

Workout Progress

Milestone 2

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Movements and Actions

ACTION 01

Forehand Drive

Low-to-high groundstroke utilizing hip-torso rotation and lead hip weight transfer.

ACTION 02

The Overhead Smash

Explosive downward strike involving trunk stretch-shortening cycle and shoulder rotators.

ACTION 03

Dink Shot & Split Step

Low-crouch Kitchen positioning with isometric lower-body control and soft paddle touch.

Fitness Goals

Goal: Increase continuous pickleball aerobic & court agility capacity from 20 to 45 minutes at a moderate-to-vigorous intensity (65–75% of Heart Rate Max) within 8 weeks, through 5 structured paddle-based workout sessions per week.

Before Measure: Resting heart rate: 78 bpm. Can currently sustain continuous low-crouch Kitchen line positioning for about 20 minutes before lower-body fatigue sets in.

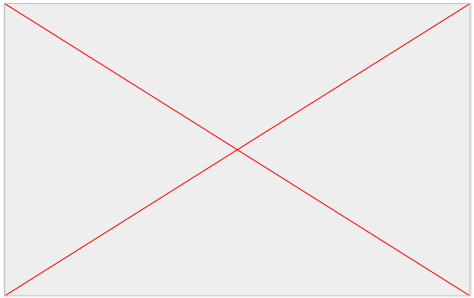
Current Measure: Pickleball court lateral shuffle & split-step endurance under 17 minutes. Resting heart rate below 73 bpm.

Frequency of Workouts: 4 sessions per week

Workout Days: Monday, Tuesday, Thursday, and Saturday

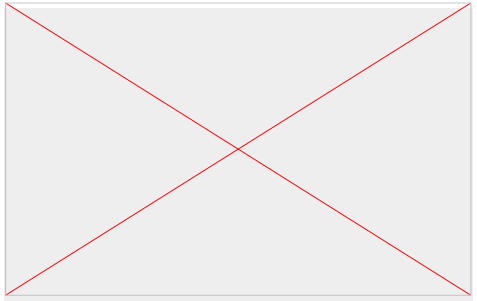
Duration of Workouts: 30 minutes total (20 minutes of dynamic bodyweight pickleball exercises with 5–minute warm-up and 5–minute cool-down)

PHYSIOLOGICAL CONSIDERATIONS



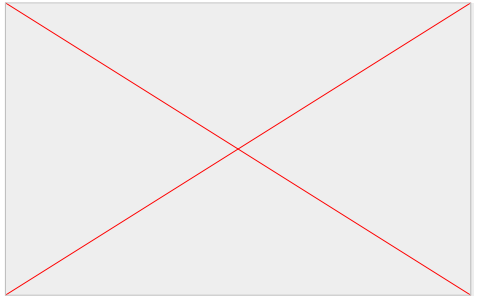
MOVEMENT / ACTION 1: FOREHAND DRIVE

Kinetic chain power transfer starting from ground reaction forces (GRF) via leg extension through trunk/hip rotational acceleration (obliques, core stabilizers). Requires shoulder horizontal adduction strength and forearm dynamic stability.



MOVEMENT / ACTION 2: THE OVERHEAD SMASH

Explosive ATP-PCr power via the stretch-shortening cycle (SSC) during chest and shoulder pre-stretch. High-velocity shoulder internal rotation (>1000°/s) and heavy eccentric deceleration demands from the rotator cuff to prevent joint impingement.



MOVEMENT / ACTION 3: DINK SHOT & SPLIT STEP

Isometric and eccentric quadriceps/gluteal strength to maintain a low center of gravity at the Kitchen line. Relies on glycolytic/aerobic capacity during prolonged crouch exchanges and fast-twitch recruitment for reaction speed.

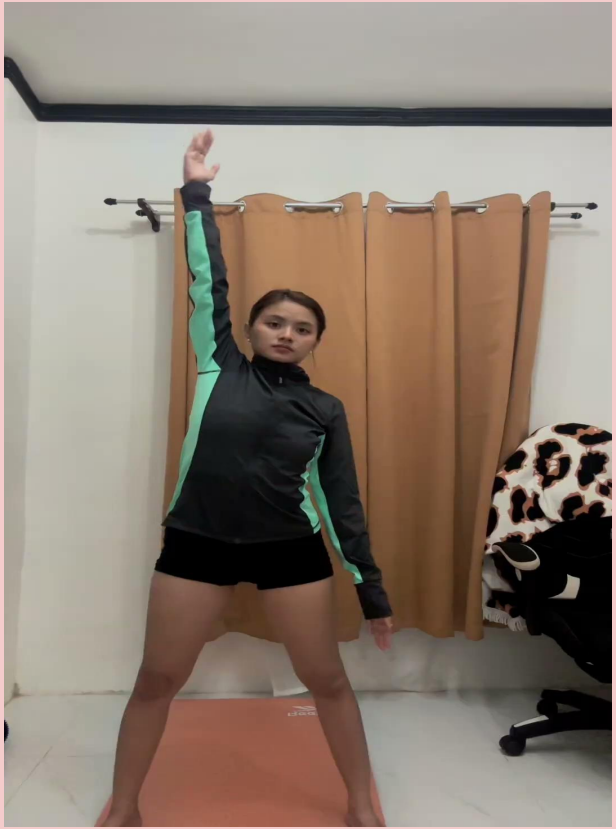
Movement/Action 1:

Forehand Drive

Bodyweight Rotational Windmill Drive

https://www.youtube.com/shorts/utEsGAg6UJU

- **Muscles Worked:**
  Obliques, rectus abdominis, gluteus medius, and anterior deltoids.
- **Steps/Tips:**
  - Stand in an athletic ready stance with feet wider than shoulder-width.
  - Pivot off back foot, drive hips forward, and sweep arms low-to-high across torso.
  - Perform 12–15 dynamic rotational shadow drive repetitions per side.
  - Focus on continuous kinetic chain rotation from hips up to shoulders.
- **Benefits or Target Individuals:**
  - Trains kinetic chain rotation and energy transfer from legs to torso without needing external equipment.



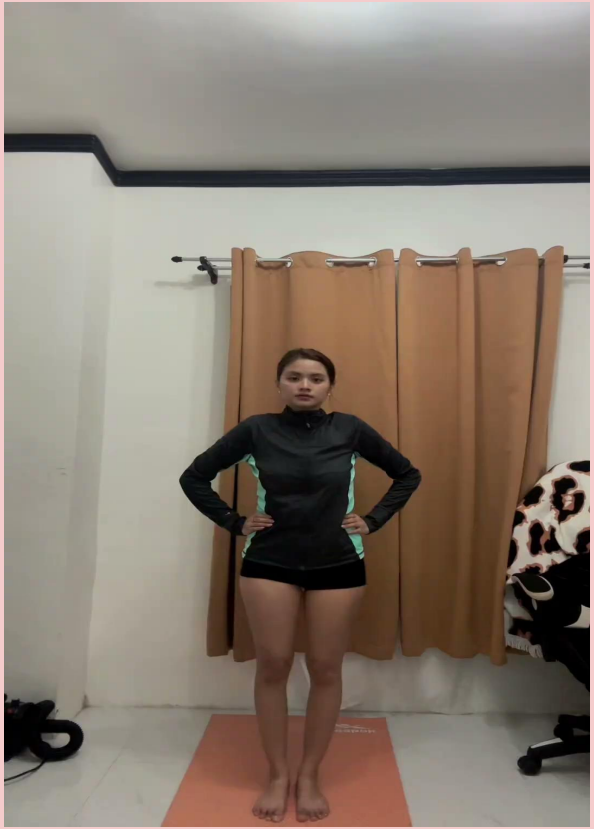
Movement/Action 1:

Forehand Drive

Bodyweight Lateral Lunge with Rotational Reach

https://www.youtube.com/shorts/VaJOo2Kb1OU

- **Muscles Worked:**
  Quadriceps, gluteus maximus, adductors, obliques, and core stabilizers.
- **Steps/Tips:**
  - Stand tall with feet hip-width apart and arms relaxed in front.
  - Step wide to the right, loading into the right hip while keeping left leg straight.
  - Reach arms across the lunging thigh to simulate groundstroke loading.
  - Push off the heel explosively to center; repeat 10–12 reps per side.
- **Benefits or Target Individuals:**
  - Replicates weight transfer and frontal-plane force absorption required during wide forehand setups.



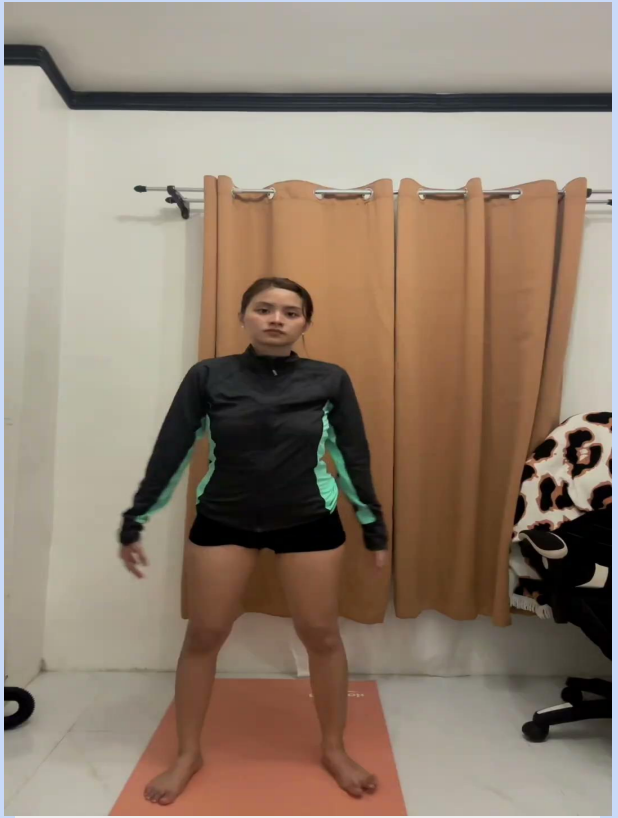
Movement/Action 2:

THE OVERHEAD SMASH

Plyometric Overhead Bodyweight Slams

https://www.youtube.com/shorts/v911k2pRgzc

- **Muscles Worked:**
  Latissimus dorsi, pectoralis major, rectus abdominis, and triceps brachii.
- **Steps/Tips:**
  - Stand in a staggered stance reaching both arms high overhead into a full pre-stretch.
  - Drive off back foot and forcefully snap arms down past hips, flexing core explosively.
  - Include a slight vertical hop at peak stretch to train plyometric ground reaction power.
  - Perform 10–12 maximum speed repetitions per set.
- **Benefits or Target Individuals:**
  - Develops stretch-shortening cycle (SSC) power and explosive core flexion required for winning overhead smashes.



Movement/Action 2:

THE OVERHEAD SMASH

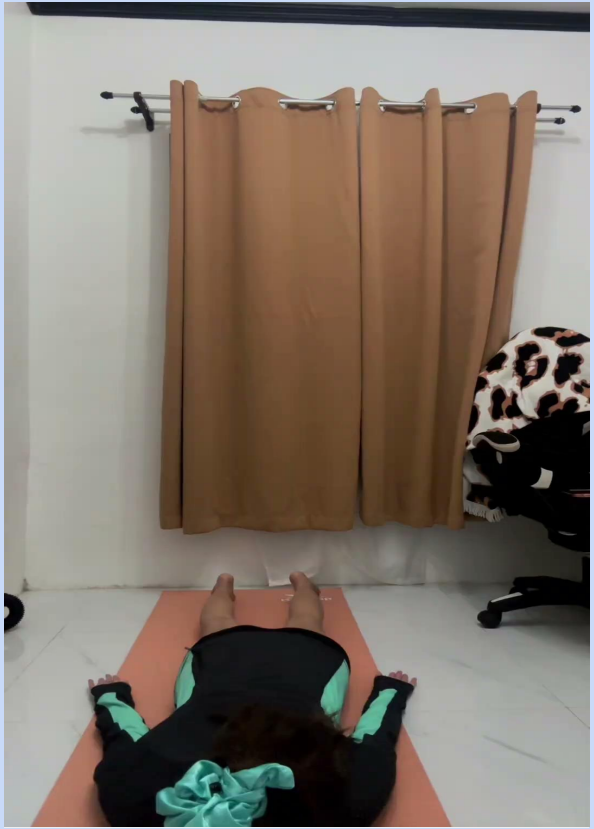
Prone Cobra Shoulder Retractions

https://www.youtube.com/shorts/v911k2pRgzc

- **Muscles Worked:**
  Infraspinatus, teres minor, rhomboids, middle/lower trapezius, and erector spinae.

- **Steps/Tips:**
  - Lie face down (prone) on floor with legs extended and forehead rested gently.
  - Squeeze shoulder blades together, lifting chest and hands slightly off mat.
  - Rotate thumbs toward the ceiling to engage external shoulder rotators.
  - Hold peak contraction for 3 seconds; perform 12–15 controlled reps.

- **Benefits or Target Individuals:**
  - Strengthens posterior rotator cuff muscles to safely decelerate high-speed overhead strikes and prevent joint impingement.



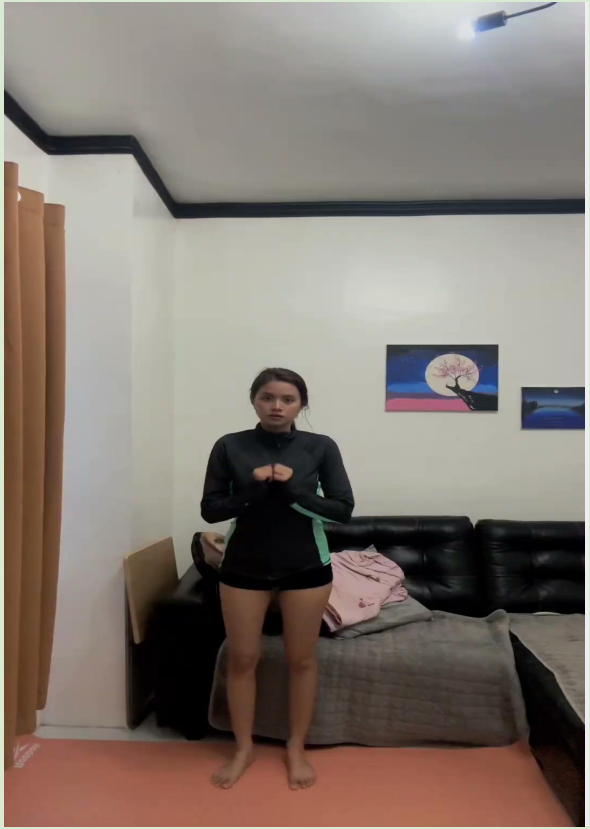
Movement/Action 3:

DINK SHOT & SPLIT STEP

Low-Crouch Lateral Shuttle Shuffle

https://www.youtube.com/shorts/nqLsCj7pgbw

- **Muscles Worked:**
  Quadriceps, gluteus medius, gastrocnemius, soleus, and hip abductors.
- **Steps/Tips:**
  - Sink into a half-squat ready stance holding hands out in front.
  - Shuffle dynamically 3 steps laterally without crossing feet over.
  - Pause briefly in balance, execute a shadow soft dink push, then shuffle back.
  - Maintain low center of gravity throughout for 30–45 seconds per set.
- **Benefits or Target Individuals:**
  - Builds lateral hip stability and lower-body endurance for sustained Kitchen-line dink exchanges.



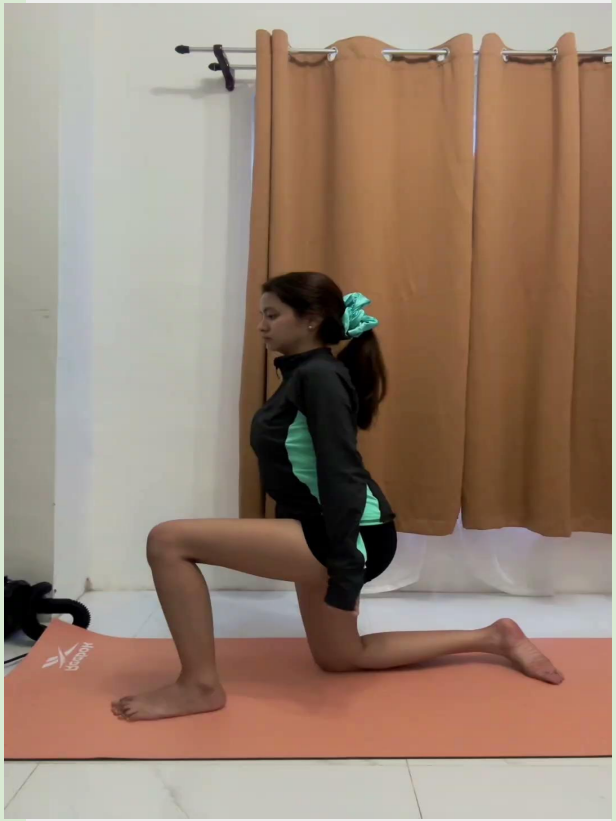
Movement/Action 2:

DINK SHOT & SPLIT STEP

Isometric Squat Hold

https://www.youtube.com/shorts/X5RfG8u8oQU

- **Muscles Worked:**
  Quadriceps, gluteal complex, calves, and core flexors.
- **Steps/Tips:**
  - Lower into a low crouch hold (60°–90° knee bend) simulating Kitchen stance.
  - Hold isometric squat stance for 5 seconds keeping chest upright.
  - Immediately reset into squat hold; complete 8–10 continuous cycles.
- **Benefits or Target Individuals:**
  - Increases isometric thigh stamina and reactive calf/ankle stiffness for rapid split-step timing.



References

Baechle, T. R., & Earle, R. W. (2008). *Essentials of Strength Training and Conditioning* (3rd ed.). Human Kinetics.

Lees, A. (2003). Science and the major racket sports: a review. *Journal of Sports Sciences*, 21(9), 707–732.

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